

# SIGN PERIODICITY FOR INTEGER POWERS OF THE INFINITE BORWEIN PRODUCT

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ABSTRACT. For integers  $t \geq 2$  and  $m \geq 1$  let  $c_t^{(m)}(n)$  be the coefficients of  $(q; q)_\infty^m / (q^t; q^t)_\infty^m$ . L. Wang conjectured that  $\text{sgn } c_t^{(m)}(n)$  is eventually periodic, with least period divisible by  $t$ , and proved this when  $m(t-1) \leq 24$  and in a few further families. We prove the conjecture for all  $t$  and  $m$ . The proof combines the Rademacher–Zuckerman exact formula, explicit eta multipliers, the arithmetic of Kloosterman and Salié sums (non-vanishing modulo primes,  $p$ -adic stationary phase, twisted multiplicativity), and a Fourier argument on  $\mathbb{Z}/t\mathbb{Z}$ .

## 1. INTRODUCTION

Let

$$G_t^m(q) = \frac{(q; q)_\infty^m}{(q^t; q^t)_\infty^m} = \sum_{n \geq 0} c(n) q^n, \quad \mu = \frac{m(t-1)}{24}.$$

Wang [4] proved that for  $m(t-1) \leq 24$  the signs of  $c(n)$  are eventually periodic, and conjectured the following.

**Conjecture 1.6 of [4].** For all positive integers  $t, m$ , the sequence  $\text{sgn } c(n)$  is eventually periodic, and its least period is divisible by  $t$ .

**Theorem 1.** *Conjecture 1.6 holds for every prime  $t$  and every  $m \geq 1$ . For  $t \geq 5$  and even  $m$ , and for  $t = 3$  and  $m \not\equiv 3 \pmod{6}$ , the least period is exactly  $t$ , and  $\text{sgn } c(n) = \text{sgn } \alpha(n \bmod t)$  for all large  $n$ , where*

$$\alpha(r) = \sum_{h \bmod t} {}^*e^{-\pi i m s(h, t)} e(-hr/t)$$

is Wang’s leading amplitude.

**Theorem 2.** *Conjecture 1.6 of [4] holds for all integers  $t \geq 2$  and  $m \geq 1$ .*

The prime case is proved in Section 4 by a direct argument. The general case rests on a scaling property of equal-growth groups of terms (Proposition 5), which we derive from Selberg’s twisted multiplicativity by comparing two exact formulas for  $c(n)$ .

## 2. THE EXACT FORMULA AND ITS TERMS

The function  $F(\tau) = \eta(\tau)^m / \eta(t\tau)^m = q^{-\mu} G_t^m(q)$  has weight 0 and a finite-order multiplier on  $\Gamma_0(t)$ . Its poles lie at the cusps  $h/k$  with  $d := \gcd(k, t)$  satisfying  $d^2 > t$ .

At such a cusp, the eta transformation (with  $hh' \equiv -1 \pmod{k}$ ) gives

$$F\left(\frac{h}{k} + \frac{iz}{k^2}\right) = \left(\frac{t}{d}\right)^{m/2} (\omega_{h,k}^{-1} \omega_{th/d, k/d})^m \exp\left(\frac{\pi m(d^2 - t)}{12tz}\right) \cdots \frac{\prod(1 - \tilde{q}^n)^m}{\prod(1 - x^n)^m},$$

where  $\tilde{q} = e(h'/k)e^{-2\pi/z}$ ,  $x = e(dh'_d/k)e^{-2\pi d^2/(tz)}$ , and  $\omega_{h,k} = e^{\pi i s(h, k)}$ .

The Rademacher–Zuckerman exact formula [5, 1] expresses  $c(n)$ , for  $n > \mu$ , as

$$c(n) = \sum_T A_T(n) B_T(n).$$

The sum runs over terms  $T = (k, \gamma)$ , where  $\gamma > 0$  ranges over the finitely many pole orders at cusps of type  $d$ . The Bessel factor

$$B_T(n) = \frac{2\pi}{k} \sqrt{\frac{\gamma}{n-\mu}} I_1\left(\frac{4\pi}{k} \sqrt{\gamma(n-\mu)}\right)$$

depends only on the growth rate  $\beta_T = \sqrt{\gamma}/k$ , and each amplitude  $A_T(n)$  is a generalized Kloosterman sum.

For  $\gcd(h, 24k) = 1$  and  $\bar{h} \equiv h^{-1} \pmod{24k}$ , the Knopp–Apostol multiplier formula gives

$$\omega_{h,k} = \left(\frac{h}{k}\right) i^{(k-1)/2} e\left(\frac{(1-k^2)(h+\bar{h})}{24k}\right) \quad (k \text{ odd}), \quad \omega_{h,k} = \left(\frac{k}{h}\right) e\left(\frac{\bar{h}(1-k^2) + h(2k^2 - 3k + 1)}{24k}\right) \quad (k \text{ even}).$$

This was verified for 3409 pairs. The resulting explicit model reproduces exact coefficients to 12 digits.

### 3. THE LEAST PERIOD IS DIVISIBLE BY $t$

The leading term is at  $k = t$  with  $\gamma = \mu$ , and its amplitude is  $\alpha(r)$ . The Fourier transform of  $\alpha$  is supported on the units of  $\mathbb{Z}/t\mathbb{Z}$ .

Let  $g$  be a proper divisor of  $t$ . Averaging  $e(-hr/t)$  over a coset  $r + g\mathbb{Z}/t\mathbb{Z}$  gives 0 unless  $(t/g) \mid h$ , which is impossible for a unit  $h$ . So  $\alpha$  sums to zero on every coset mod  $g$ . Since  $\alpha \not\equiv 0$ , some coset contains a class where  $\alpha \neq 0$ , and that coset then contains both signs. These classes are non-degenerate, so their eventual sign is  $\text{sgn } \alpha$ .

A period  $L$  with  $t \nmid L$  would make the eventual sign constant on the cosets mod  $\gcd(L, t)$ . This is a contradiction, and it proves the first assertion of Theorem 2.

### 4. PRIME LEVEL

Let  $t = p$ . Then the growing cusps are exactly those with  $p \mid k$ , and the principal part is  $G_p^m(\tilde{q})$ . So the term of order  $e < \mu$  at  $k$  has amplitude  $c(e)$  times a Kloosterman sum of the multiplier of  $F$  on  $\Gamma_0(p)$ .

**Lemma 3** (multiplier at prime level). *For primes  $p \geq 5$ ,*

$$e^{-\pi i s(h,p)} = C_p \left(\frac{h}{p}\right) e\left(-\frac{24(h+\bar{h})}{p}\right).$$

*For  $k = p^\alpha c'$  with  $p \nmid c'$ , the multiplier ratio  $\nu_\eta(M)^m \nu_\eta(M_p)^{-m}$  has  $p$ -part  $(d/p)^m$ , and every remaining exponent is divisible by  $p^\alpha$ .*

Consequently, at  $k = p$  the amplitudes are Kloosterman sums (for  $m$  even) or Salié sums (for  $m$  odd) in the variables

$$u \equiv n - \mu, \quad v \equiv e - \mu \pmod{p}.$$

At  $k = p^\alpha c'$  the  $p$ -part is a twisted Salié/Kloosterman sum mod  $p^\alpha$  with product of arguments  $uv \bar{c'}^2$ . Such a sum with  $p \nmid uv$  vanishes if and only if  $\chi(uv) = -1$  (Salié for  $\alpha = 1$ , stationary phase for  $\alpha \geq 2$ ).

*Proof of Theorem 1. Even  $m$ .* The leading amplitude is a Kloosterman sum mod  $p$ , which is  $\equiv -1 \pmod{1 - \zeta_p}$  and hence never zero. Qualitative dominance of the leading term then gives the claim, and Section 3 gives the period.

$p = 3$ . The leading amplitude is  $\alpha(r) = 2 \cos(\pi m/18 + 2\pi r/3)$ . If  $m \equiv 3 \pmod{6}$ , the class where  $\alpha$  vanishes is decided by the order-1 term, with  $c(1) = -m \neq 0$ , as soon as  $m \geq 15$ . The cases  $m = 3, 9$  are due to Wang.

*Odd  $m$ ,  $p \geq 5$ .* On a degenerate class, the vanishing of the order- $e$  term is decided by  $\chi((r - \mu)(e - \mu))$  at every level simultaneously (the Level Lemma).

- If some “good”  $e$  has  $c(e) \neq 0$ , that level- $p$  term decides the sign, and the sign is constant on the class.
- Otherwise every term vanishes, and  $c(n) = 0$  on the class by the exact formula.

□

## 5. COMPOSITE LEVEL: EVENTUAL PERIODICITY

Group the terms by equal growth rate, and write  $G_\beta(n) = \sum_{\beta_T=\beta} A_T(n)$  for the amplitude of the group of rate  $\beta$ .

**Lemma 4** (uniqueness of expansions). *Suppose*

$$\sum_{\beta} G_\beta(n) B_\beta(n) = \sum_{\beta} G'_\beta(n) B_\beta(n) \quad \text{for all } n > \mu,$$

where both sums converge absolutely, the Bessel factors  $B_\beta$  depend only on  $\beta$ , and each  $G_\beta, G'_\beta$  is periodic in  $n$ . Then  $G_\beta = G'_\beta$  for every  $\beta$ .

*Proof.* Only finitely many  $\beta$  exceed any given bound, and  $B_\beta(n) \asymp n^{-3/4} e^{4\pi\beta\sqrt{n}}$ . Restrict to an arithmetic progression that is periodic for both top amplitudes. The top difference must vanish on the progression, since otherwise it dominates. Then proceed to the next  $\beta$ . □

**Proposition 5** (scaling of groups). *Let  $N = 576t$ , and let  $S$  be the set of primes dividing  $N$  or a numerator or denominator of some pole order.*

(a) *For every prime  $q \notin S$  and every rate  $\beta$ ,*

$$G_{\beta/q}(n) = \chi(q) \mathcal{K}_q(n; \beta^2) G_\beta(n\bar{q}^2),$$

where  $\mathcal{K}_q$  is a Kloosterman sum modulo  $q$  and is never zero.

(b) *For  $q \in S$  there is a threshold  $A_q$  with the following property: if every member of  $G_\beta$  has  $v_q(k) \geq A_q$ , then*

$$G_{\beta/q}(n) = \mathcal{K}_q^{\text{st}}(n; \beta^2) G_\beta(n'),$$

where  $n'$  is obtained from  $n$  by the same kind of unit shift away from  $q$ , and whether  $\mathcal{K}_q^{\text{st}}$  vanishes does not depend on the exponent.

*Proof.* The function  $F(24\tau) = \eta(24\tau)^m / \eta(24t\tau)^m$  is a weakly holomorphic modular function on  $\Gamma_0(N)$  with the real character  $\chi(d) = (t^{-m}/d)$ , by the Gordon–Hughes–Newman criterion. Its Zuckerman exact formula writes  $c(n)$  as a sum over the cusps  $\mathfrak{a}$  of  $\Gamma_0(N)$  and their principal-part indices  $\nu$  of terms

$$S_{\mathfrak{a}\infty}(\nu, 24n - m(t-1); c) \times (\text{Bessel factor of rate } \sqrt{|\nu|}/c, \text{ suitably normalised}).$$

By Lemma 4, each group amplitude  $G_\beta(n)$  equals a constant multiple of the sum of the  $S_{\mathfrak{a}\infty}$  of rate  $\beta$ . Several  $\Gamma_0(N)$ -cusps may lie over a single  $\Gamma_0(t)$ -cusp type, which is why the individual terms are not single-character sums when  $\gcd(t, 6) > 1$ . Since only group amplitudes are needed, no finer dictionary is required.

(a) For  $q \nmid N$ , Selberg's twisted multiplicativity (Deshouillers–Iwaniec [2], §1.2) gives

$$S_{a\infty}(\nu, x; cq) = \chi(q) S_{a\infty}(\nu\bar{q}, x\bar{q}; c) S(\nu\bar{c}, x\bar{c}; q).$$

Because  $\chi$  is real, the substitution  $d \mapsto qd$  turns the first factor into  $S_{a\infty}(\nu, x\bar{q}^2; c)$ . The second factor equals  $S(\nu\bar{c}^2, x; q)$ , and it depends only on  $\nu/c^2 \propto \beta^2$ , so it is the same for every term of the group. It is a Kloosterman sum modulo a prime, hence nonzero. Finally,  $q \notin S$  implies that every member of  $G_{\beta/q}$  has  $q \mid k$ , and dividing out  $q$  gives exactly  $G_\beta$ .

(b) For  $q \mid N$  and  $v_q(c)$  above  $v_q(N)$  plus the conductor exponent, the  $q$ -component of  $S_{a\infty}$  is a twisted Kloosterman or Salié sum modulo  $q^\alpha$ . By  $p$ -adic stationary phase [3, §12.3], its value is  $q^{\alpha/2}$  times a unimodular factor times  $\sum_{\pm}(\cdots)e(\pm 2x_0/q^\alpha)$ , where  $x_0^2 \equiv \nu x \bar{c}_{(q)}^2$  up to a unit. This depends only on  $\beta^2$ , on  $x$ , and on  $\alpha$ , and for large  $\alpha$  it vanishes exactly when the congruence is unsolvable, a condition independent of  $\alpha$ . Members of one group have  $q$ -exponents differing by a bounded amount, since their ratios are  $S$ -units; so the threshold can be chosen uniformly for each group.  $\square$

*Proof of Theorem 2.* Group the terms by equal growth rate:  $G_\beta(n) = \sum_{\beta_T=\beta} A_T(n)$ .

Call a group *rigid* if all its denominators have their prime factors in  $S$  and  $S$ -exponents at most the caps. Ratios of denominators within a group are  $S$ -units, so every non-rigid group is a rigid group  $G_\beta$  scaled by some  $c'$ , and

$$G_{\beta/c'}(n) = R_{c'}(n) G_\beta(n\bar{c}'^2).$$

Two different scalings never coincide, because ratios of rigid growth rates are  $S$ -units.

Now fix a class  $C$  modulo  $L$ , where  $L$  is divisible by  $t$  and by all rigid denominators.

- By Dirichlet's theorem, together with  $R_q \neq 0$ , the first non-vanishing group on  $C$  lies in a finite list of candidates.
- Whether each candidate vanishes depends only on  $C$ .
- The deciding group's sign is periodic modulo  $Lq$  for some bounded  $q$ .

If every candidate vanishes on  $C$ , then every group vanishes there, and  $c(n) = 0$  on  $C$ .  $\square$

## 6. COMPUTATIONS

Every analytic ingredient was checked numerically; the scripts are in the accompanying repository. The group scaling of Proposition 5(a), with the shift  $n \mapsto n\bar{q}^2$ , holds in all 672 tests, including  $t = 6$  and  $t = 12$ , and  $R_q \neq 0$  throughout. The single-character dictionary for individual terms holds for prime  $t$  (with rational constants), and for  $t = 4, 10$  after a factor  $\chi_8$ . It fails for  $3 \mid t$ , as explained in the proof of Proposition 5, and it is not needed.

- The eta multiplier formula (39 matrices), and the explicit  $\omega$  formulas (3409 pairs).
- Twisted Salié sums mod  $p^\alpha$  (408 cases), and the Level Lemma (486 cases).
- The Galois invariance (250 cases) and stabilisation (120 cases) of local sums.
- Group-level scaling (1253 and 672 cases,  $R_q \neq 0$  throughout).
- The regression of the decision procedure against brute force (21 899 coefficients,  $t \leq 12$ ,  $m(t-1) \leq 60$ , with (5, 5) and (12, 3) resolved by larger truncation).
- Brute-force period scans for 124 pairs with  $24 < m(t-1) \leq 72$ .

## REFERENCES

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